Aluminium-cell ground pump

Evolved for catalytic efficiency — register entry H38

kronspice simulation report · 2026-11-07 · seed 1, NSGA-II 40 × 25

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Summary

The negative Cockcroft–Walton ground pump of *photon_electron* was rebuilt with its rectifiers replaced by aluminium–water valve cells (the Pollak/Nodon electrolytic rectifier of *sparkgap_diode*). A genetic search (NSGA-II) then designed the cells and the drive so that electrons are pumped from ground onto a negative bank with as little energy as possible. Two layouts were evolved: plate stacks, with each cell in its own tank, and concentric pipes, with one tank per rectifier.

- **Best result:** η = 0.857 at −321 V and 7.8 mA from 2.9 W, using one cell of six 13 mm² plates formed at 545 V and driven at 471 Hz. It delivers 1.66×10^{16} electrons per joule, against the e·|V| ideal of 1.94×10^{16}. A single aluminium rod in an iron cup does slightly better: η = 0.869 at −197 V.
- **The baseline is lost in the tank, not the rectifiers.** At 10 V and 100 kHz, 98–99 % of the drive heats the tank coil, so η is 0.0006 with silicon diodes and 0.00002 with aluminium cells. Evolution moves to small tanks at hundreds of hertz.
- **The oxide film's capacitance is the stack's main barrier.** Every efficient design uses the least aluminium area and the highest forming voltage it can, with cells in series.
- **η falls with voltage:** 0.5 at −3.4 kV and 0.15 at −10 kV, because the coil's share of the loss grows again.
- **Silicon diodes are still better,** by 0.01–0.09 in η on every evolved design. The 2 % varactor effect of the oxide lowers η by up to 0.013, and it is far too small for parametric (2f) pumping.
- **No design goes under the e·|V| floor.** The ledger enforces this as a test, so the result is a measure of how close a real circuit gets to the floor, not a way around it.

What “catalytic efficiency” means

Lifting one electron from ground to a bank at −|V| costs at least e·|V|, whatever the circuit. That is a floor, and no resonance or barrier lowering goes under it. What a circuit *can* change is how much more than e·|V| it spends. The fitness is therefore

- **$\eta = P_{load} / P_{in}$** in steady state. The bank voltage must drift less than 1 % over the measuring window, so energy still going into capacitors is not counted as output.
- Two further objectives: the bank voltage and the load current.
- Two constraints: a stack never sees more than 90 % of its holding voltage in reverse, and the bank must be in steady state.

η = 1 would mean the drive pays only e·|V| per electron. Each design's input is split into the load, coil resistance, conducting rectifiers (electrolysis onset plus electrolyte), leaking rectifiers and the rest. The parts sum to the input within 1 %.

The aluminium valve stack

A stack is *Ns* cells in series × *Np* plates in parallel, with plate area *A*, a water gap and a forming voltage V_{form}:

property	value	from
holds (reverse)	Ns · Vform	oxide formed at Vform, ≤ 700 V
forward onset	Ns · 2 V	electrolysis per cell
forward resistance	Ns · ρ · gap / (A · Np), ρ = 0.1 Ω·m	water electrolyte
reverse leakage	A·Np · j0 · exp(β·E), 0.1 A/m² at Vform	Güntherschulze–Betz, $\beta = 3\times10^{-8}$ m/V, 1.4 nm/V

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property	value	from
above Vform	sparks (2 Ω per cell)	breakdown
film capacitance	εr 9.5 · ε0 · A·Np / (1.4 nm/V · Vform) / Ns	formed oxide
varactor	C(v) = C0 · (1 + κ·|v|/hold), κ = 2 %	fixed-thickness barrier oxide

Cells in series raise the voltage the stack holds and divide its film capacitance. Plates in parallel raise the current and multiply the film capacitance. The 11 genes are drive frequency (50 Hz–100 kHz) and amplitude (1–50 V), tank capacitor, stages (1–8), ladder capacitors, load, cells in series (1–10), plates in parallel (1–20), plate area (10 mm²–100 cm²), gap (1–20 mm) and forming voltage (50–700 V). The film is evolved as a linear capacitor (κ = 0), which runs 16× faster. Every Pareto design is then re-measured with κ = 2 % and again with silicon diodes.

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Results: plate stacks

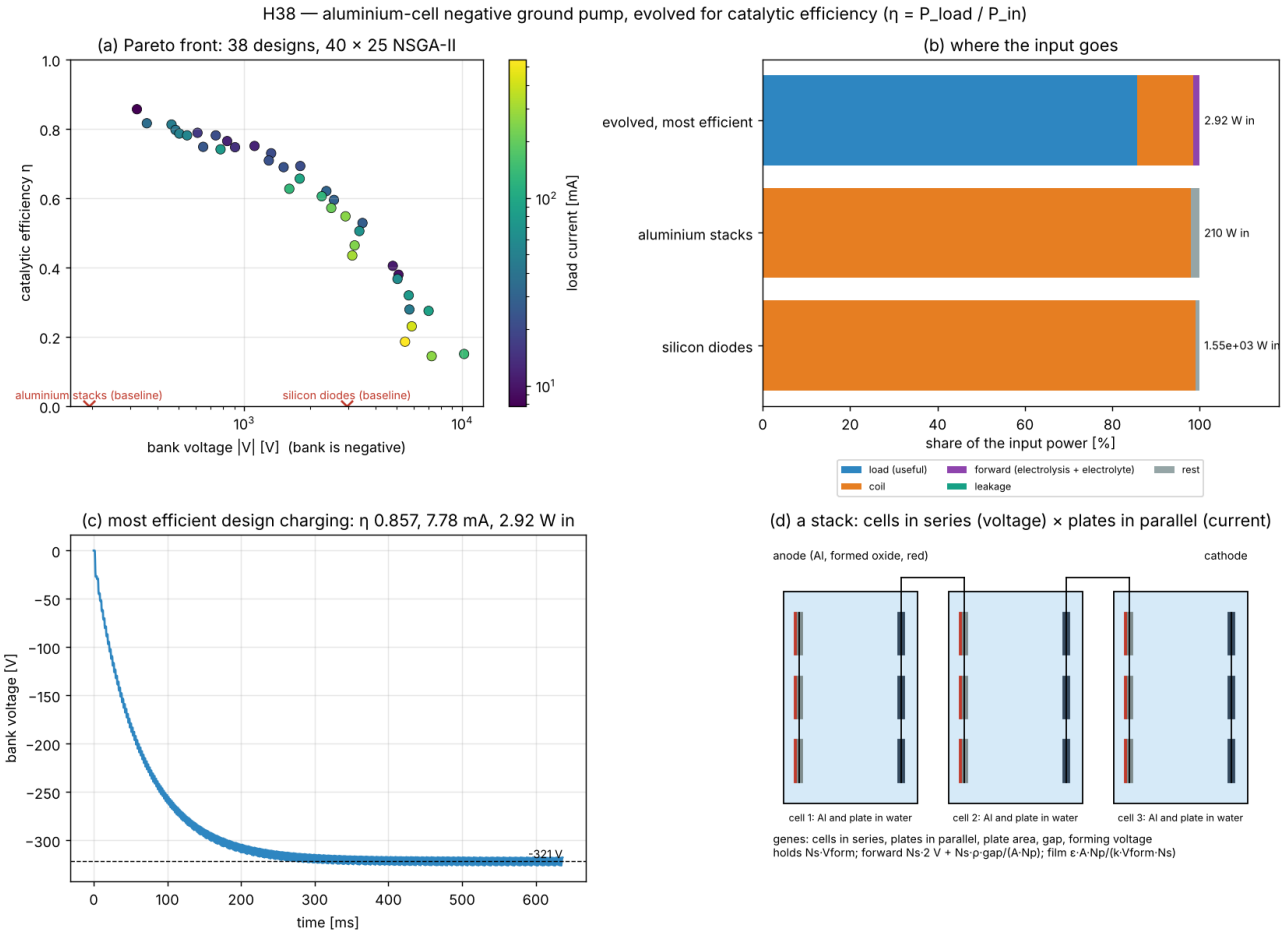


Figure 1. (a) The Pareto front, with the baselines at the bottom. (b) Where the input goes. (c) The most efficient design charging its bank from rest. (d) A multi-plate stack.

design	bank	load	input	η	coil	forward
baseline, silicon diodes (10 V, 100 kHz, 1 µF tank)	−2953 V	0.30 mA	1.55 kW	0.0006	98.9 %	0 %
baseline, aluminium stacks	−194 V	0.02 mA	210 W	0.00002	98.0 %	0 %
evolved, most efficient (1 cell × 6 plates)	**−321 V**	**7.8 mA**	**2.9 W**	**0.857**	**12.8 %**	**1.4 %**
evolved, 5 cells × 3 plates	−357 V	36 mA	15.6 W	0.816		
evolved, 8 cells × 5 plates, 2 stages	−1113 V	12 mA	17.9 W	0.751		
evolved, 7 cells × 10 plates, 3 stages	−3372 V	69 mA	459 W	0.505		
evolved, 7 cells × 8 plates, 3 stages	−10.2 kV	140 mA	9.5 kW	0.150		

Most efficient design. One cell of six 13.2 mm² plates with a 7.9 mm gap, formed at 545 V, in one stage. It is driven at 471 Hz and 50 V into a 0.25 µF tank. The stack holds 545 V; its forward path is 2.0 V plus 9.96 Ω, and its film is 0.00875 µF. The tank swings 179 V, and the largest reverse voltage is 60 % of the oxide's rating. The electrolysis onset alone would allow $\eta \le |V|/(|V| + 4\ V) = 0.988$; most of what is lost goes to the coil.

What the evolution found

- **The baseline's loss is the tank.** A 1 µF tank at 100 kHz circulates hundreds of amps to hold its swing. Evolution moves to 0.2–1 µF tanks at hundreds of hertz to a few kilohertz. There the coil takes 13–20 % on the designs above η = 0.75.
- **The film capacitance is the main barrier.** At 100 kHz the baseline stack's 0.05 µF film all but shorts it, the high-frequency failure the 1911 *Britannica* describes. Efficient designs push plate area to 13–60 mm² and forming voltage to 545–700 V, and stack cells in series.

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- **Conduction costs little.** The 2 V per-cell onset and the water's resistance together take 0.4–9 % of the input.
- **Leakage is negligible** once reverse voltage is kept under 90 % of the forming voltage.
- **Voltage and current cost efficiency.** Above about −1 kV, designs need more stages and bigger tanks. The coil's share then grows again to 50–84 %.
- **Silicon is still better** by 0.01–0.09 in η. The aluminium stack works, but its film and its onset voltage are costs, not a catalyst.

The parametric route (2f pumping)

Driving the tank at 2f through the film's C(v) grows an oscillation at f only if the capacitance swing $m = \Delta C/C$ exceeds $2/Q$ (the Mathieu threshold). The formed oxide gives $m \leq 2$ % even when swung to its forming voltage, so it would need $Q > 100$. The coil's Q of 50 needs $m > 4$ %. The oxide is too linear to pump parametrically; a varactor diode, whose capacitance changes several-fold, could.

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One tank per rectifier: concentric pipes

In this layout an aluminium rod sits at the centre, pipes surround it and the tank wall is outside. Each annulus of water between two neighbouring pipes is one cell. Two wirings were evolved:

- **Parallel (bus bars).** The pipes alternate aluminium and iron, and each metal goes to its own bus bar, so every annulus is the same valve between the same two buses. Neighbouring annuli sit at the same voltage, so the pipe ends need no seals. The tank holds one forming voltage.
- **Series (clad pipes).** Each pipe is aluminium outside and iron inside, and the current crosses every annulus in turn, so the tank holds N forming voltages. Neighbouring annuli differ by one cell voltage, so every pipe bottom must be sealed. In **graded series**, the outer pipes' aluminium is masked so that every annulus has the same area.

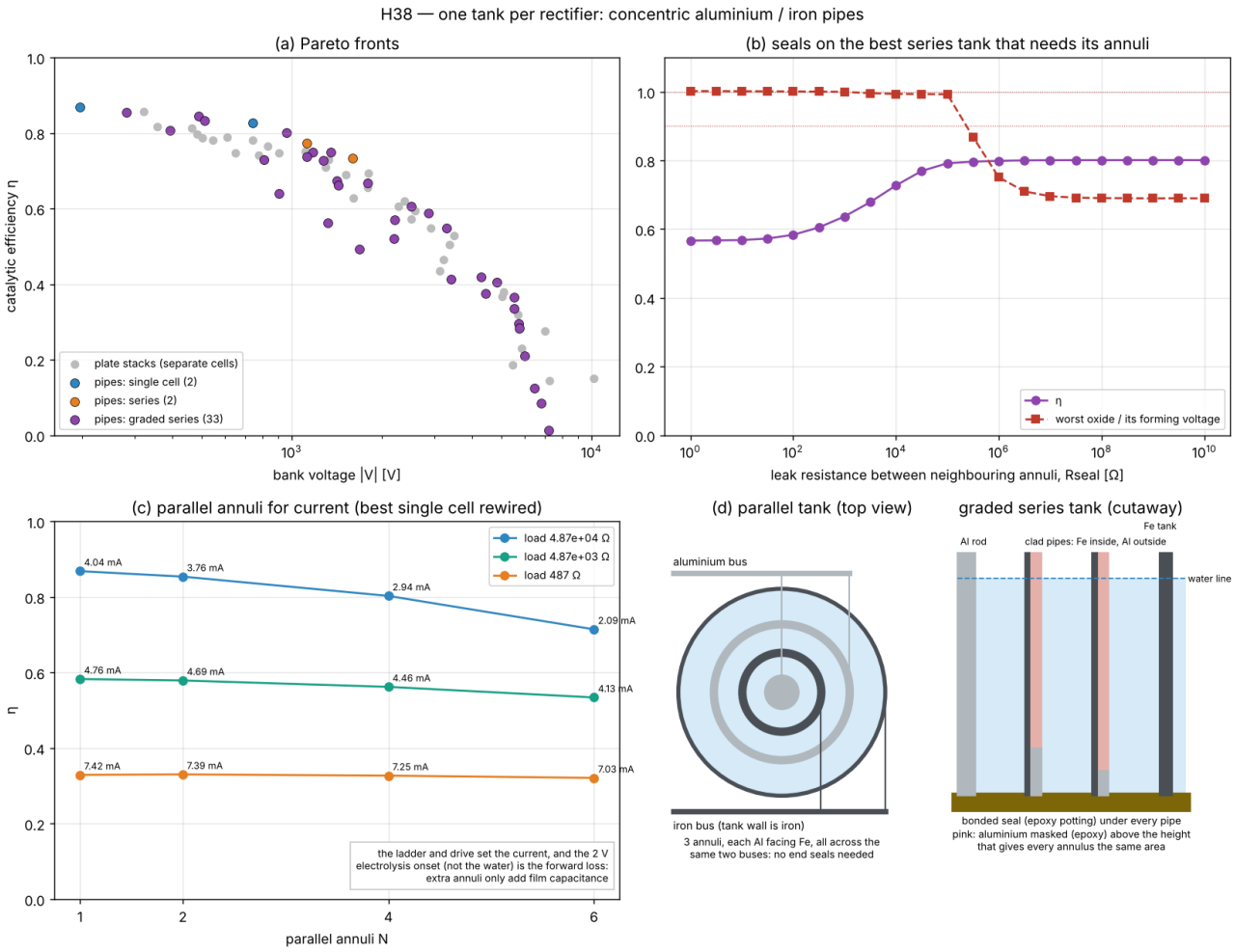


Figure 2. (a) Pipe fronts over the plate-stack front. (b) Seal leakage on the best series tank that needs its annuli. (c) Parallel annuli for current. (d) The two tank layouts.

tank	bank	load	input	η
single cell: Al rod Ø3.9 mm in an iron tank Ø7 mm, 5 mm wetted, formed at 414 V	**−197 V**	**4.0 mA**	**0.92 W**	**0.869**
graded series, 2 annuli, tank Ø18 mm	−282 V	6.7 mA	2.2 W	0.856
graded series, 2 annuli, tank Ø10 mm	−959 V	31 mA	37 W	0.801
graded series, 8 annuli, tank Ø42 mm, 3 stages	−2.5 kV	13 mA	53 W	0.606
graded series, 5 annuli, tank Ø57 mm, 4 stages	−4.8 kV	19 mA	228 W	0.405

- **Pipes match plates.** The pipe front lies on the plate front: η 0.87 against 0.86 at a few hundred volts, and 0.6 against 0.6 at 2.5 kV. One tank of up to 10 annuli replaces up to 10 separate cells.
- **The less aluminium, the better.** The wetted length stays at its 5 mm minimum on almost every design. The best rectifier is a thimble: one aluminium rod in an iron cup.

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- **Parallel annuli do not help at these currents.** Rewiring the best single cell as 2, 4 or 6 parallel annuli lowers η: 0.869, 0.854, 0.803 and 0.715. The current is set by the ladder and the drive, and the forward loss is the 2 V onset, not the water. Parallel annuli pay off only near 1 A.
- **Plain series tanks load the inner annulus.** The AC voltage divides as 1/C, so in a test design the annuli take 92, 16 and 9 % of their forming voltage. In the graded tank they share it equally (41/41/41 %), and graded series tanks fill 33 of the 37 places on the front.
- **Seals matter only when the tank needs its annuli.** The −959 V tank (1.38 forming voltages) keeps 99 % of its η only with at least about 0.3 MΩ between annuli. Under a 5 mm-wide seal, that means a water film thinner than about 70 nm: a bonded, potted seal, not a gasket. Pipes merely standing on the floor (about 13 Ω) spark the inner annulus, and η falls to 0.57.

Limits of the model

- The cell constants are representative literature values (water resistivity, Güntherschulze–Betz β, oxide growth, leakage at forming), not measurements on a particular cell. A bench cell should be characterised for forward onset, leakage against voltage and film capacitance, and these constants replaced.
- The film does not age or re-form during a run. A real oxide loses its holding voltage when idle and re-forms with current.
- The coil has a fixed Q of 50. A better coil, or a lower-loss tank, would move the whole front up.
- Electrolysis gas, heating and changes in the electrolyte are not modelled. Foam over the pipe tops could bridge annuli like a leaky seal.
- Forward current density is not limited. The high-current designs reach about 1 kA/m² on small aluminium faces, where gassing would be heavy.

Reproducing the results

`go run ./examples/al_cell_pump` runs both studies, two 40 × 25 evolutions taking about 15 min on 2 CPUs. `-study pipes` runs only the pipe study, and `-data DIR` writes the CSVs for the figures. `go test ./examples/al_cell_pump` takes about 6 s and checks, among other things, that electrons per joule never exceed the e·|V| ideal. The full output is in `run_output.txt`.

Status in the register: **bounded**. The standard prediction was computed, and nothing goes under e·|V|.

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